www.tntextbooks.in

11.6.2 Structure of Stomata

The epidermis of leaves and green stems possess many small pores called **stomata**. The length and breadth of stomata is about 10-40μ and 3-10μ respectively. Mature leaves contain between 50 and 500 stomata per $mm^{2.}$ Stomata are made up of two **guard cells**, special semi-lunar or kidney-shaped living epidermal cells in the epidermis. Guard cells are attached to surrounding epidermal cells known as **subsidiary cell**s or **accessory cells**. The guard cells are joined together at each end but they are free to separate to form a pore between them. The inner wall of the guard cell is thicker than the outer wall (Figure 11.14). The stoma opens to the interior into a cavity called **sub-stomatal cavity** which remains connected with the intercellular spaces.

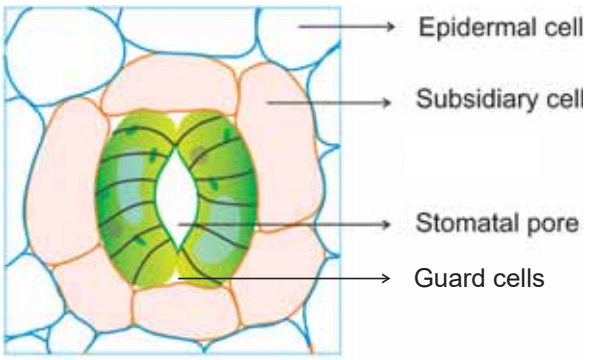


Figure 11.14: Structure of Stomata

11.6.3 Mechanism of Stomatal Movement

Stomatal movements are regulated by the change of turgor pressure in guard cells. When water enters the guard cell, it swells and its unevenly thickened walls stretch up resulting in the opening of stomata. This is due to concave non-elastic nature of inner wall pulled away from each other and stretching of the convex elastic natured outer wall of guard cell.

Different theories have been proposed regarding opening and closing of stomata. The important theories of stomatal movement are as follows,

1. *Theory of Photosynthesis in guard cells*
2. *Starch – Sugar interconversion theory*
3. *Active potassium transport ion concept*

1. Theory of Photosynthesis in guard cells

Von Mohl (1856) observed that stomata open in light and close in the night. According to him, chloroplasts present in the guard cells photosynthesize in the presence of light resulting in the production of carbohydrate (Sugar) which increases osmotic pressure in guard cells. It leads to the entry of water from other cell and stomatal aperture opens. The above process *vice versa* in night leads to closure of stomata.

Demerits

1. Chloroplast of guard cells is poorly developed and incapable of performing photosynthesis.
2. The guard cells already possess much amount of stored sugars.

2. Starch – Sugar Interconversion theory

i. According to **Lloyd** (1908), turgidity of guard cell depends on interconversion, of starch and sugar. It was supported by **Loftfield** (1921) as he found guard cells containing sugar during the daytime when they are open and starch during the night when they are closed.

ii. **Sayre** (1920) observed that the opening and closing of stomata depends upon change in pH of guard cells. According to him stomata open at high pH during day time and become closed at low pH at night. Utilization of CO_2